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Characterisation of antenna substrate properties using surrogate-based optimisation

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Abstract: In the current study, the authors present a technique to characterise the permittivity and loss tangent of an antenna substrate. In this method, an error function is formulated with measured data and a set of simulation data on the antenna's impedance and bandwidth. Subsequently, a surrogate-based optimisation code is used to intelligently search for the substrate parameters that minimise the error function. The validity of the method is demonstrated by experiments on wearable patch antennas with felt substrates. The results show that the method is effective in retrieving the permittivity and loss tangent of the antenna substrate using the antenna itself as the test fixture. Thus, there is no need to prepare additional resonators or transmission-line fixtures as in conventional substrate characterisation methods.

1 Introduction

It is crucial to know the substrate material parameters prior to designing an antenna, as they significantly affect the antenna figures of merit (e.g. resonance frequency, bandwidth (BW), efficiency and physical size). In many cases, the substrate's relative permittivity (ε_r) and loss tangent ($\tan\delta$) are provided by the manufacturer, and an antenna engineer refers to them to carry out a proper antenna design.

In recent years, new materials have been increasingly used, and their material parameters are often not clearly identified. For example, wearable antennas employ off-the-shelf fabric or foam materials as substrates, as they provide a comfortable wearing experience at a low manufacturing cost [1–6]. However, their material properties are not readily available, and their values are less homogeneous. In this regard, a customised method of characterising new substrate materials is required.

The most widely used substrate characterisation technique is the transmission-line method [7, 8]. In this method, the scattering parameters (*S*-parameters) are first measured, and a microstrip line is printed over the substrate of interest. Then, the material parameters are extracted using closed-form equations derived from the fundamental transmission-line theory. In these equations, ε_r and $\tan\delta$ are expressed in terms of the *S*-parameters under the assumption of the quasi-transverse electromagnetic wave mode propagation along the microstrip line length. Thus, its accuracy can be significantly degraded by higher-order mode propagation. Moreover, the half-open structure of the microstrip line causes inevitable radiation loss that can mislead the $\tan\delta$ result.

As alternatives to the restrictive closed-form equations, new approaches have attempted to estimate material parameters by comparing measured data to a set of simulation data [9–11]. More specifically, measured data of the antenna figures of merit are compared with a set of simulated data obtained by varying ε_r and $\tan\delta$ in the simulation model. Meanwhile, an optimisation tool is used to search for the value that minimises the difference between the two (e.g. error function). One efficient optimisation technique is surrogate-based optimisation (SBO) [9, 10]. This technique precisely locates the minimum of the error function based on intelligently chosen data points. Therefore it is useful for solving complicated non-linear problems with less data collection and computation effort. For antenna substrate characterisation, the error function can be formulated by one or more antenna figures of merit. In [10], for instance, the magnitude of the antenna reflection coefficient ($|S_{11}|$) and the radiation efficiency (e_{cd}) are used.

In this paper, we formulate the error function using the antenna impedance (Z) and BW to improve the accuracy and facilitate the data-collection process. The impedance $Z = R + jX$ (Ω) is more intuitive in representing ε_r and $\tan\delta$ rather than $|S_{11}|$. Furthermore, the radiation efficiency, used in previous studies, requires additional measurement set-ups (e.g. anechoic chamber or Wheeler cap), making the process complicated and expensive. The measured radiation efficiency is also somewhat vulnerable to testing and calibration error, which may mislead the final characterisation results. On the other hand, Z and BW can be conveniently measured with any network analyser provided that operates up to the antenna's resonant frequency.

In the following, the effectiveness of the proposed method is examined by means of full-wave simulations and measurements. Section 2 presents the overall characterisation process including the error function calculation and the SBO practice. Section 3 presents studies based on full-wave simulations. In particular, different combinations of error functions are compared and analysed. Section 4 demonstrates the measurement and characterisation results of patch antennas designed on felt substrates of different thicknesses. The results show that the proposed error function together with the SBO technique is useful in characterising ε_r and $\tan\delta$ of the substrate without the need of additional transmission-line fixtures and radiation efficiency measurements.

2 Substrate characterisation process

Changing ε_r and $\tan\delta$ of the substrate affects Z and BW of an antenna. This leads to the inverse problem of determining the substrate's ε_r and $\tan\delta$ from the antenna's Z and BW. Fig. 1 shows the overall substrate characterisation process. The measured $\tilde{Z}$ and $\widetilde{\mathrm{BW}}$ are obtained from the antenna fabricated on the substrate of interest. Concurrently, in a full-wave simulation tool (e.g. ANSYS high-frequency structure simulator (HFSS)), an antenna model with identical geometry to the measurement is designed, and a set of simulated Z and BW values is collected by assigning different ε_r and $\tan\delta$ values in the antenna substrate.

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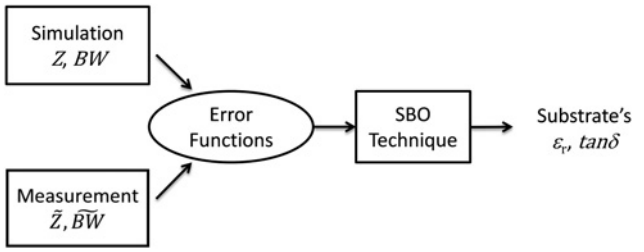


Fig. 1 *Flowchart of the substrate characterisation process*

These simulated data are compared with the measured data by an error function via the SBO technique. More specifically, we use the MATLAB SUrrogate MOdeling (SUMO) toolbox from Ghent University, Belgium [12, 13]. This toolbox provides a kriging model of the error function with the given simulation and measured data, and then it efficiently searches for the ε_r and tan δ values that minimise the error function.

2.1 Error function

The error function describes the discrepancy between the simulation data and the measured data, in that it should be carefully designed to improve the computational efficiency and accuracy of the final results. A total of four different error functions are examined in this paper. All of them are combinations of resistance (R), reactance (X) and BW as follows

$$\mathrm{EF}_1 = \frac{1}{n}\sum_{i=1}^{n}(R_i - \widetilde{R}_l)^2 \tag{1}$$

$$\mathrm{EF}_2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \widetilde{X}_l)^2 \tag{2}$$

$$\mathrm{EF}_3 = |\mathrm{BW} - \widetilde{\mathrm{BW}}| \tag{3}$$

$$\mathrm{EF}_4 = \frac{1}{n}\sum_{i=1}^{n}(R_i - \widetilde{R}_l)^2 + a \cdot |\mathrm{BW} - \widetilde{\mathrm{BW}}| \tag{4}$$

where n is the number of frequency points and R_i/X_i/BW and $\widetilde{R}_l/\widetilde{X}_l/\widetilde{\mathrm{BW}}$ correspond to the simulation and measured data, respectively. EF_1 and EF_2 represent the mean squared error of R and X, respectively, over the frequency range of observation, whereas EF_3 is the absolute subtraction of the BW. EF_4 is the combination of R and BW. Here, a weighting factor, a, is included in the BW term to keep the proportion of the two different parameters in the same order.

2.2 Surrogate-based optimisation

As mentioned, the SUMO toolbox [13] is used to generate the kriging models to precisely locate the error function minimum. The input to the toolbox is a single scattered data file in the form of six column matrices. For instance, two columns are occupied with the optimisation variables (ε_r and tan δ), and the other four are occupied with the error functions ($\mathrm{EF}_{1,2,3,4}$).

At the beginning of the optimisation process, Latin hypercube design [14] is used to select a total of 14 initial search points, including 4 corner points and 10 points in between. The very first kriging model is generated based on this. Subsequently, an adaptive sampling strategy called expected improvement (EI) is used to update the next search points. The updating process continues until the sample budget runs out. In the following simulation and measurement studies, it is found that the EI strategy quickly locates the error function minimum with a budget of <60 sample points.

3 Validation by simulation

To validate the substrate characterisation process, a simulation study was performed by considering the full-wave simulation data of a reference antenna as the measured data (Fig. 1). Fig. 2 illustrates the rectangular patch antenna used for this paper. The antenna is fed by a coaxial probe and designed to resonate at $f_r = 2.44$ GHz. Table 1 lists the antenna dimensional parameters that were carried out by the patch antenna design equations found elsewhere [15]. The conductive patch and ground are homogeneous copper foil with a conductivity of $\sigma = 5.8 \times 10^7$ S/m. The antenna substrate is a high-frequency laminate RO4350B (i.e. FR4) with $\varepsilon_r = 4.4$ and tan $\delta = 0.018$. These values are expected to be retrieved at the end of the substrate characterisation process.

A set of an antenna's S_{11} data was collected using the same simulation model as above, but by varying ε_r and tan δ of the substrate. Next, 11 sampled points were assigned in the range of $\varepsilon_r \in [4, 5]$ and tan $\delta \in [0.001, 0.05]$, resulting in a total of $11 \times 11 = 121$ samples. A computer code was developed to collect the mass amount of data in HFSS automatically. Fig. 3 shows the S_{11} data of the reference antenna with FR4 on top of the sampled S_{11} data. As expected, the reference data falls in the sampled data. The remaining task is to determine the reference antenna's ε_r and tan δ efficiently and accurately based on the error function and subsequent SBO process.

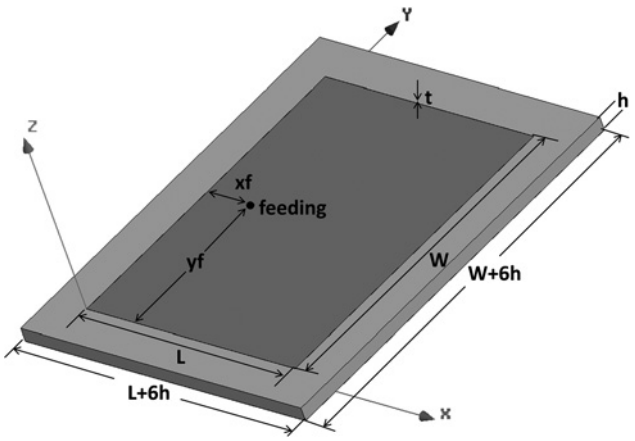


Fig. 2 *Geometry of the antenna model*

Table 1 Dimensions of the antenna model

f_r, GHz	t, mm	h, mm	W, mm	L, mm	xf, mm	yf, mm
2.44	0.018	1.6	38	28	6	17

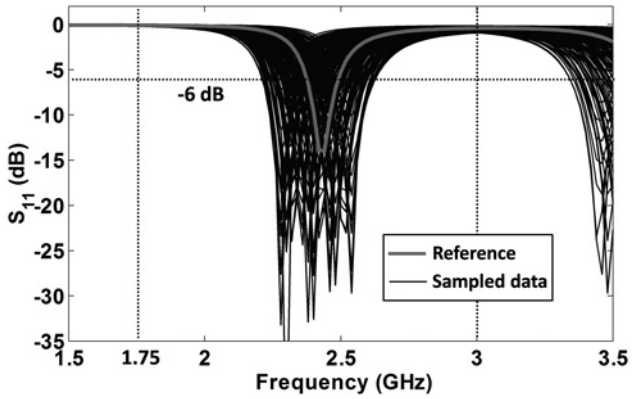


Fig. 3 *Sampled and reference data of simulated S_{11}*

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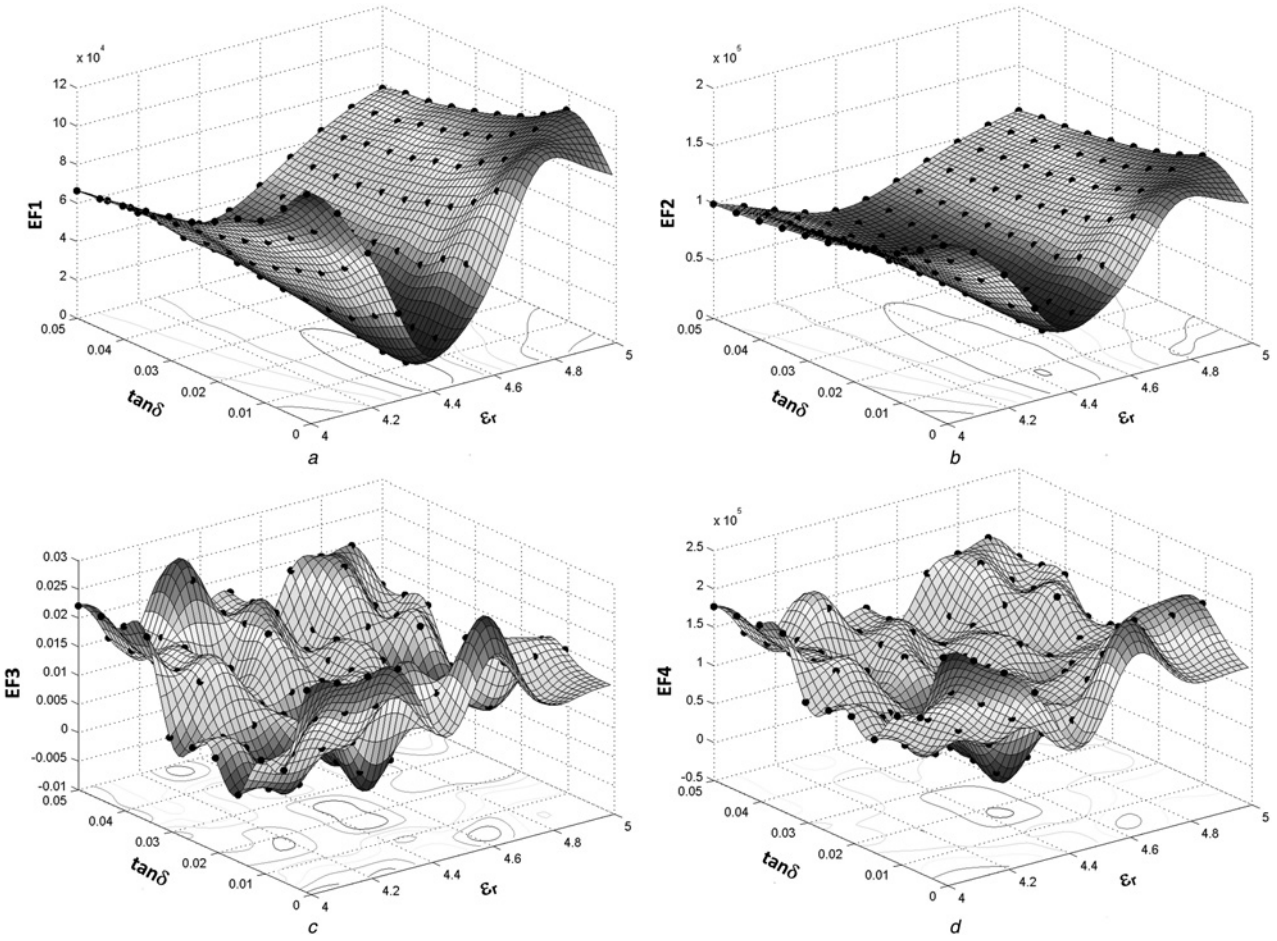


Fig. 4 *Optimisation results of simulation data*

a Using EF_1
b Using EF_2
c Using EF_3
d Using EF_4

Table 2 Retrieved ε_r and tan δ for each error function

	REF	EF_1	EF_2	EF_3	EF_4
ε_r	4.4	4.396	4.431	4.451	4.444
tan δ	0.018	0.008	0.008	0.017	0.018

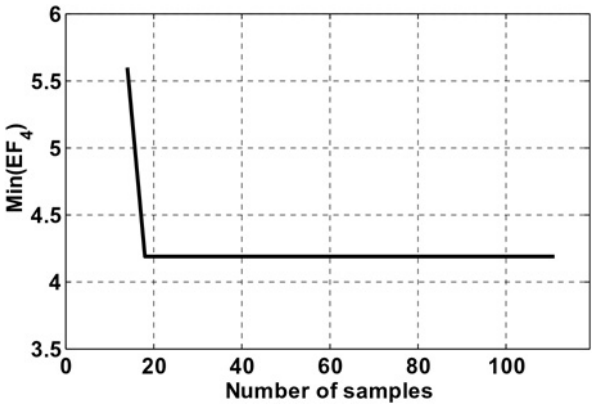


Fig. 5 *EF_4 minimum against number of samples*

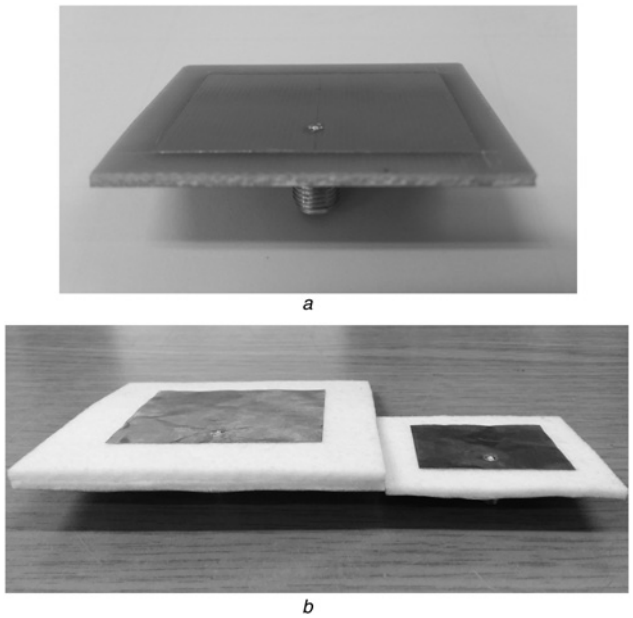


Fig. 6 *Pictures of fabricated antennas with*

a FR4
b Felt

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Table 3 Dimensions of fabricated antenna

	Antenna substrate	f_r, GHz	h, mm	W, mm	L, mm	xf, mm	yf, mm
1	FR4	2.44	1.6	38	28	6	17
2	felt	4.37	2.4	38	28	6	17
3	felt	2.59	4.8	58.3	48.8	7	24.4

It is worth noting that choosing the frequency range for the error function parameters, Z and BW, is important. As can be seen in Fig. 3, some of the sampled S_{11} data exhibit more than one resonance. Empirical studies showed that the accuracy of the resulting ε_r and $\tan\delta$ degraded as multiple resonances were included in the input data of the SBO process. In this context, the Z data in the frequency range of $f \in [1.75, 3]$ GHz was only considered as the input data to exclude the second resonance (see Fig. 3). For the BW data, the $S_{11} < -6$ dB BW criterion was used. It was found a more rigorous criterion such as $S_{11} < -10$ dB tended to degrade the accuracy, as some of the sampled data were not resonating under -10 dB. The -6 dB BW was automatically identified by a computer code and sent to the SUMO toolbox to facilitate the process.

The error functions consisting of the above Z and BW data were loaded in the SUMO toolbox as described in Section 2. Figs. 4*a–d* show the optimisation results (i.e. final kriging models) using, $EF_{1,2,3,4}$, respectively. In Figs. 4*a* and *b*, it can be seen that the kriging models built from only R and X of the antenna impedance have similar distributions and a clear valley vertical to the ε_r-axis. This implies that R and X are suitable antenna figures of merit to evaluate the substrate's ε_r, as they are sensitive to the change in ε_r. Fig. 4*c* shows the generated kriging model when only BW is used in the error function (i.e. EF_3). Opposite to the previous model, a clear valley is exhibited vertical to the $\tan\delta$-axis, implying that BW is more sensitive to the change in $\tan\delta$ than ε_r. In fact, $\tan\delta$ highly affects the quality factor determining the resonance BW. Otherwise, the resonance frequency is more dependent on the impedance influenced by ε_r.

Given these observations, we can expect that EF_4 formulated by both R and BW is sensitive to both ε_r and $\tan\delta$. This is displayed in Fig. 4*d*. We note that the weighting factor of $a = 5 \times 10^6$ is used in (4) to guarantee that R and BW are in the same order. As a result, a distinctive optimum is located on the plane with the values of $\varepsilon_r = 4.444$ and $\tan\delta = 0.017$, which are close to the reference data $\varepsilon_r = 4.4$ and $\tan\delta = 0.018$. Table 2 summarises the retrieved ε_r and $\tan\delta$ results. It is observed that $EF_{1,2}$ are less accurate in determining $\tan\delta$, whereas EF_3 have more errors in the retrieved ε_r.

Fig. 5 shows the minimum value of EF_4 against the number of samples used in the SBO process. The minimum value dramatically reduces as the number of samples reaches 18 among the total 121 samples, but then no improvement is made by adding more samples. This verifies the computational efficiency of the EI strategy adopted in the SUMO toolbox [14].

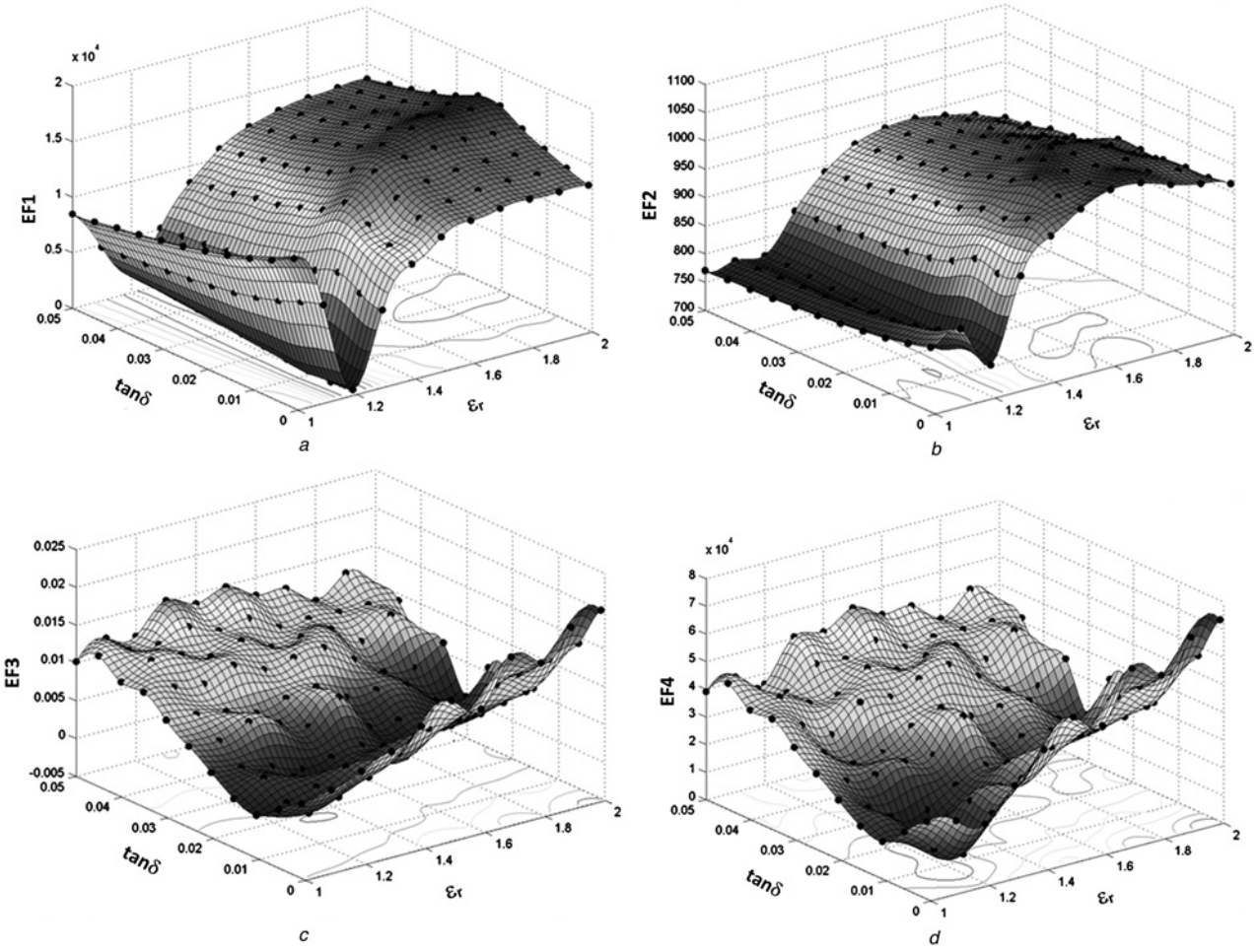


Fig. 7 *Optimisation results of measured data with felt substrate (Antenna 2)*

a Using EF_1
b Using EF_2
c Using EF_3
d Using EF_4

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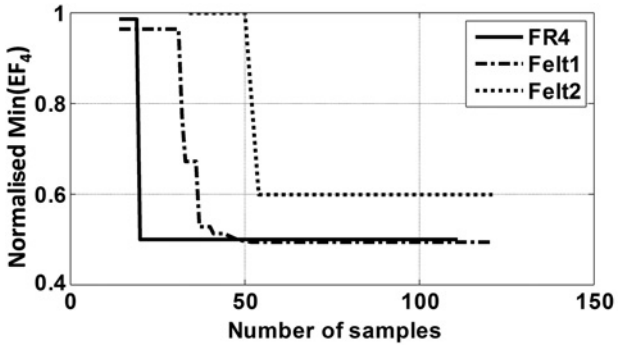


Fig. 8 *Comparison of normalised EF_4 minimum against number of samples for three tested antennas*

4 Measurement results

Having validated the proposed method using simulations, we proceed to characterise antenna substrates based on measurements. Fig. 6 shows three antennas fabricated on (a) FR4 and (b) felt substrates. For the latter, two antennas of different dimensions were prepared with the same felt substrate. This is to determine whether the same material parameters can be retrieved even if the antennas with different resonant frequencies and BWs are under evaluation. Table 3 shows the dimensions of the test antennas. All parameters correspond to those in Fig. 2.

With the fabricated antennas, Z and BW were measured by a network analyser (Anritsu MS2038C) and then processed in the SUMO toolbox. As each antenna resonated at different frequencies, the Z data were considered in the frequency ranges of $f_1 \in [1.75,\ 3]$ GHz, $f_2 \in [3,\ 5]$ GHz and $f_3 \in [1.5,\ 3.2]$ GHz for Antennas 1 (FR4), 2 (thin felt) and 3 (thick felt), respectively. The BW data continued to use the $S_{11} < -6$ dB criterion.

Figs. 7*a*–(*d*) depict the optimisation results in the case of Antenna 2. The three-dimensional kriging model plots show similar distributions to the simulation study: a clear valley vertical to the ε_r-axis for EF_1 and EF_2 and a valley vertical to the tan δ-axis for EF_3. In addition, Fig. 7*d* obtained from EF_4 has a sharp minimum point at $\varepsilon_r = 1.242$ and $\tan\delta = 0.018$. The characterisation results of all three antennas are summarised in Table 4. The maximum error from the reference data is 6.9% for ε_r and 10% for tan δ in the case of the felt substrate antennas. It is also shown that the retrieved results from the two felt substrate antennas with different thicknesses are identical for tan δ and exhibit <2.6% difference for ε_r, implying that the method is feasible to characterise substrate material parameters regardless of the antenna's size, resonant frequency and BW. For the FR4 antenna, the calculated errors are 0.6% for ε_r and 5.5% for tan δ.

The normalised minimum values of EF_4 for the three fabricated antennas are compared in Fig. 8. The EF_4 minimum is converged as the number of samples reaches 20, 48 and 54 for the FR4, thin felt and thick felt antennas, respectively. It takes more time for the EI strategy to explore the area of the optimised minimum for the thick felt antenna, but the required sample number is still <60.

Table 4 Retrieved ε_r and tan δ using EF_4

	Antenna substrate	Reference [11]		Measured results	
		ε_r	tan δ	ε_r	tan δ
1	FR_4	4.4	0.018	4.372	0.017
2	felt	1.3	0.02	1.242	0.018
3	felt	1.3	0.02	1.211	0.018

5 Conclusions

In this paper, we proposed a property characterisation technique for antenna substrates based on SBO. A set of full-wave simulation data and error functions with antenna figures of merit were prepared as the input of the SBO algorithm. Four different error functions were evaluated, and the results showed that the error function formulated by Z and BW was capable of retrieving the substrate ε_r and tan δ with high accuracy.

Compared with the ordinary characterisation method, the proposed method offers in-situ characterisation of the antenna substrate. That is, the characterisation can be performed right on the fabricated antenna. Thus, preparing a different test fixture is not necessary. Furthermore, we expect that the proposed technique will make data collection more convenient than the previous SBO-based substrate characterisation technique, as there is no need to acquire the radiation efficiency of the antenna.

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7 References

1 Hall, P.S., Hao, Y.: 'Antennas and propagation for body-centric wireless communications' (Artech House, Norwood, MA, USA, 2012, 2nd edn.)
2 Tsolis, A., Whittow, W.G., Alexandridis, A.A., *et al.*: 'Embroidery and related manufacturing techniques for wearable antennas: challenges and opportunities', *Electronics*, 2014, **3**, (2), pp. 314–338
3 Ouyang, Y., Chappell, W.J.: 'High frequency properties of electro-textiles for wearable antenna applications', *IEEE Trans. Antennas Propag.*, 2008, **56**, (2), pp. 381–389
4 Kennedy, T.F., Fink, P.W., Chu, A.W., *et al.*: 'Body-worn e-textile antennas: the good, the low-mass, and the conformal', *IEEE Trans. Antennas Propag.*, 2009, **57**, (4), pp. 910–918
5 Wang, Z., Zhang, L., Psychoudakis, D., *et al.*: 'Embroidered multiband body-worn antenna for GSM/PCS/WLAN communications', *IEEE Trans. Antennas Propag.*, 2014, **62**, (6), pp. 3321–3329
6 Kim, S., Kwon, K., Choi, J.: 'Design of a miniaturized high-isolation diversity antenna for wearable WBAN applications', *J. Electromagn. Eng. Sci.*, 2013, **13**, (1), pp. 28–33
7 Baker-Jarvis, J., Vanzura, E.J., Kissick, W.A.: 'Improved technique for determining complex permittivity with transmission/reflection method', *IEEE Trans. Microw. Theory Tech.*, 1990, **38**, (8), pp. 1096–1103
8 Cottet, D., Gryzb, J., Kirstein, T., *et al.*: 'Electrical characterization of textile transmission lines', *IEEE Trans. Adv. Packag.*, 2003, **26**, (2), pp. 182–190
9 Couckuyt, I., Declercq, F., Dhaene, T., *et al.*: 'Surrogate-based infill optimization applied to electromagnetic problems', *Int. J. RF Microw. Comput. Aided Eng.*, 2010, **20**, (5), pp. 492–501
10 Declercq, F., Couckuyt, I., Rogier, H.: 'Environmental high frequency characterization of fabrics based on a novel surrogate modeling antenna technique', *IEEE Trans. Antennas Propag.*, 2013, **61**, (10), pp. 5200–5213
11 Wu, Y.Q., Tang, Z.X., Zang, B., *et al.*: 'Permeability measurement of ferromagnetic materials in microwave frequency range using support vector machine regression', *Prog. Electromagn. Res. B*, 2008, **70**, pp. 247–256
12 Gorissen, D., Couckuyt, I., Demeester, P., *et al.*: 'A surrogate modelling and adaptive sampling toolbox for computer based design', *J. Mach. Learn. Res.*, 2010, **11**, pp. 2051–2055
13 'SUMO SUrrogate MOdeling Lab.'. Available at http://www.sumo.intec.ugent.be, accessed April 2014
14 Dam, E.R., Van Husslage, B.G.M., den Hertog, D., *et al.*: 'Maximin Latin hypercube designs in two dimensions', *Oper. Res.*, 2007, **55**, pp. 158–169
15 Balanis, C.A.: 'Antenna theory: analysis and design' (Wiley, New York, NY, USA, 2005, 3rd edn.)

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